

SAKSHAM MRIG

Tel: +91 9717380635 | Email: mrigsksum@gmail.com | [Linkedin](#)

EDUCATION

INDRAPRASTHA INSTITUTE OF INFORMATION TECHNOLOGY, DELHI
BTech in Computer Science and Applied Mathematics | CGPA – 8.68/10

New Delhi, India
2019 – 2023

WORK EXPERIENCE

ATLASSIAN | *Software company that develops collaboration tools ([Jira](#) & [Confluence](#)) for developers and PMs* **Bengaluru, India**
Associate Data Scientist **Jun 2023 – Present**

- Optimized allocation of cloud-infrastructure for enterprise customers to enhance user experience, through reduced latency
 - Analyzed user-behaviour, cloud instance shape and cloud-infrastructure/geo-location segments for current customers; performed a cohort analysis study using Amplitude to understand drivers of performance issues faced by end-users
 - Engineered a Linear Regression model to predict growth in weekly cloud traffic across different regions; combined the forecasts with performance guardrails and predicted costs to identify the optimal mix of distributed Storage and VM systems
 - Collaborated with product managers to optimize allocation of appropriate cloud-infrastructure on a per-customer basis, reducing cloud residency cost for **200+** enterprise customers by **~8%** and decreasing average page response time by **15%**
- Built a cloud trial-to-sales propensity (Conversion-Likelihood) model for improved audience targeting
 - Calculated historical “Cloud-Trial” adoption likelihood, conversion rates from past sales data and user activity statistics, and further passed these features into a Bayesian Model to determine the likelihood of adoption for prospective customers
 - Assisted the Demand Generation Team to identify customers with a higher potential upside (**+43%** conversion rate); highlighted under-utilisation of trials in specific cohorts and unexplored customer segments to drive migration sales
- Forecasted data storage and operational costs to strategize the pricing and packaging of a new Backup/Restore Product
 - Developed an Multiple Linear Regression (MLR) model using past product adoption and monthly usage data of 300K+ enterprise and SMB customers to predict future operational costs through different phases of a new Backup/Restore Product
 - Utilized these projections, along with competitor analysis, to build an optimized monthly usage-based pricing and packaging strategy; saved more than **\$1.5k** /customer/month in potentially unused AWS storage costs for Atlassian
- Established a fully automated source-control framework for legacy Data Pipelines developed by the team
 - Facilitated automatic syncing between the code repository and the data warehouse employing Bitbucket Pipelines, AWS and Apache Hive API; migrated **50+** legacy ETLs to the revamped architecture, making subsequent collaborations efficient
 - Built a Data-Quality Recommendation system, using Anthropic’s Claude 3.5, fine-tuned on internal SQL tables to suggest and implement fundamental data quality checks and save **10+Hr/Week** of developer time
- Data-Driven Performance Optimization of Atlassian’s Cloud Migration Assistant Suite
 - Designed and executed A/B tests for Cloud Migration Assistants’ performance and user-experience updates; utilized statistical significance tests to evaluate effectiveness of new features in improving performance metrics like Page Response Time

Data Science Intern

May – Aug 2022

- Tailored Cloud Migration Strategies Using Decision Trees and Machine Learning
 - Recommended the best cloud migration strategy for customers, by mapping business logic and guardrails used by Enterprise Partners into a Decision Tree, including factors such as Data complexity, Tentative downtime and other Technical specs
 - Suggested the best options tailored for smaller customers by using past migration data from SMB Customers to train a Random Forest Classifier; achieved an Average Precision score of **0.70** and saved hours for our Enterprise Partners
- Engineered a dashboard employing a pipeline using PySpark, Databricks and Plotly that gathers post migration-to-cloud survey responses, to identify and rank topics, using the Empath Topic Modelling algorithm, to gauge key drivers of customer satisfaction
- Synthesized data from the migration survey responses into **20+** insights, such as ranking and rewarding partners based on feedback from customers, time spent during the various phases of the migration, etc

GOOGLE SUMMER OF CODE | *Google-funded global program for students to contribute to Open Source Software* **Remote**
Software Developer Intern | [Github](#) **Apr – Aug 2020**

- Refactored the legacy monolithic codebase to follow an MVC (Model-View-Controller) architecture by implementing an API layer, using Javascript, to make the method calls asynchronous and core interpreter efficient; streamlined codebase for future development
- Added 10+ new blocks in the visual programming interface including a “Virtual Piano” and an “Oscilloscope” block built using Javascript, Web MIDI API and Tone.js to enable MIDI compatibility, benefiting over **2,000+** instructors and learners
- Resolved **71** github-bugs and delivered feature requests like an online project sharing platform for students built on a MongoDB database; using NLP, to improve in-app block search by **24%**; launched an offline mobile version for 100+ students in Peru

PUBLICATIONS

Leveraging Intra and Inter Modality Relationship for Multimodal Fake News Detection

Apr 2022

Authors: Shivangi Singhal, Tanisha Pandey, Saksham Mrig, Ponnurangam Kumaraguru, Rajiv Ratn Shah
WWW’ 22: The ACM Web Conference | Lyon, France | Github

- Built a Fake News Detection framework trained on two publicly available multimodal (Images and Text) datasets - “Verifying Multimedia Use - Challenge” and “Weibo Fake News”, exploiting a blend of intra and inter modality relationships
- Presented a novel architecture that identifies and suppresses information from weaker modalities and extracts relevant information from strong modality on a per-sample basis; outperformed existing methodologies by **3.05%** (accuracy) and **4.53%** (F1-score)
- Captured the intra-modality relationship in images via a bottom-up attention approach using an R-CNN framework

ACHIEVEMENTS

Finalist, Mathematical Contest in Modelling (MCM/ICM)

Feb 2023

- Represented IIITD in the annual modelling competition organized by COMAP: Consortium for Mathematics and its Applications; ranked **3rd in India** and within the **top 3% globally** among 25,000 teams
- Developed a framework that can estimate the average number of tries required to complete the NY Times WORDLE puzzle
- Combined a word difficulty predictor, built using XGBoost classifier trained on historical words and their difficulty, with a Dirichlet Regression Model that can model the number of tries to successfully complete the puzzle; achieved a brier score of <0.1

1st Place, HackPSU | Github

Mar 2021

- Won first position among **515** international teams at HackPSU (organized by Penn State University); also won the Nittany Ai Alliance “Ai for Good” Challenge, winning a total sum of **\$1000**
- Built a chrome-extension based personal productivity and companion app that tracks browsing activity and reminds users whenever they are going off-topic or procrastinating, while also reminding users at set intervals to take short breaks
- Used the Microsoft Azure Facial Emotion model to track user’s facial expressions while browsing; generated a daily analysis report of their browsing activity alongside the emotions captured from their face, helping user’s maintain mental well-being

ACADEMIC RESEARCH AND PROJECTS

E-Commerce Fraud Detection [*Collaboration with Flipkart, a leading E-Commerce Company*]

Jan – May 2022

- Collaborated with Flipkart to diminish the unwanted logistics cost for e-commerce companies through in-transit cancellations and returns by bad actors using a machine learning model trained on past purchase trends as well as buyers’ social media features
- Built a Buyer Social Re-Identification framework that successfully searches and filters for prospective profiles from social media platforms using identifiers from sale records; augmented a dataset of 25000 product sales by adding buyers’ social media profiles
- Trained a Random Forest Classifier on past purchase trends/features and social features extracted from LinkedIn and Twitter, achieving a goodness of **1633.7 bps** and **88.8%** precision, which is currently in use by Flipkart to proactively identify fraudsters

Enhanced Pythontutor Tool for Advanced ByteCode Visualization | Github | Online Tool URL

Aug – Dec 2022

- Developed an advanced version of Pythontutor.com, a program visualization tool widely-used in CS education, with additional functionality to analyze ByteCode execution and Evaluation Stack operations at the heart of Python runtime
- Extracted the Internal Symbols, AST and Evaluation Stack from CPython’s runtime at each step of ByteCode execution and displayed it in an interactive NodeJs WebApp for easy visual learning and debugging during the execution of Python Bytecode
- Deployed the website for **200+** students studying “Intro to Programming” and “Compilers” courses at IIIT Delhi

Safe Exploration in Deep Reinforcement Learning

Jan – May 2023

- Designed and developed a RL Agent that ensures compliance with critical constraints during the learning phase, applicable to fields like autonomous vehicles and data-center cooling systems
- Utilized a state-based action correction layer to enhance standard deep reinforcement learning algorithms, such as DDPG and TD3, ensuring safe and effective decision-making
- Evaluated the effectiveness of our Agent on Ball-3d and Spaceship Openai gym environments via faster convergence; observed **10-15%** higher cumulative reward than other alternatives, while following a zero-violation policy

Detecting Causes of Mental Health Issues in Code-Mixed Social Media Posts | Github

Aug – Dec 2022

- Scraped and augmented a dataset of 12,000 Reddit posts from top mental health subreddits; translated this data into Hindi/English code-mixed using a pretrained multilingual hi-en version of mBART to test the Multilingual robustness of the model
- Identified underlying causes of mental illness by analysing the user’s social media posts and training a Sentence-BERT model on the code-mixed data along with the CAMS dataset; gained **+5%** in accuracy over alternative benchmark models

POSITIONS OF RESPONSIBILITY

President: Byld (*Software development club @ IIIT Delhi*)

Aug 2022 – Feb 2023

- Led a team of 7 Students to organize a 2-Day technical hackathon with talks from industry leaders on exciting new fields for a cohort of **200+** student attendees and **70** participating teams
- Facilitated the pairing and outreach of a mentor-mentee cohort of **30+** freshman/sophomore students interested in upskilling technically; resulted in 3-Google Summer of Code, 2-Outreachy and 1-FOSSAsia internship acceptances

TECHNICAL SKILLS

Programming: R, Python, C/C++, Java, Ocaml

Data Science: SQL, Tableau, MATLAB, Apache Spark, Pandas, NumPy, Seaborn

ML Frameworks:, Scikit-Learn, PyTorch, Keras

ADDITIONAL INFORMATION

Community Engagement: Participated in trash clean-up activities in community parks, Bengaluru (Goodera, May 2024); volunteered in local government schools to teach elementary mathematics and serve mid-day meals, Bengaluru (KV DRDO School, July 2023)

Interests: Officially ranked 29th (by the SRFi) in U17 Junior Squash; Led the IIITD Squash team to inter-college victory in 2023; enjoy playing Keyboard (Grade 3, Trinity College London); “Developing Your Musicianship” Specialization (Berklee 2020)

Online Courses: Neural Networks and Deep Learning (Coursera, 2020); Haskell MOOC (University of Helsinki, 2023)